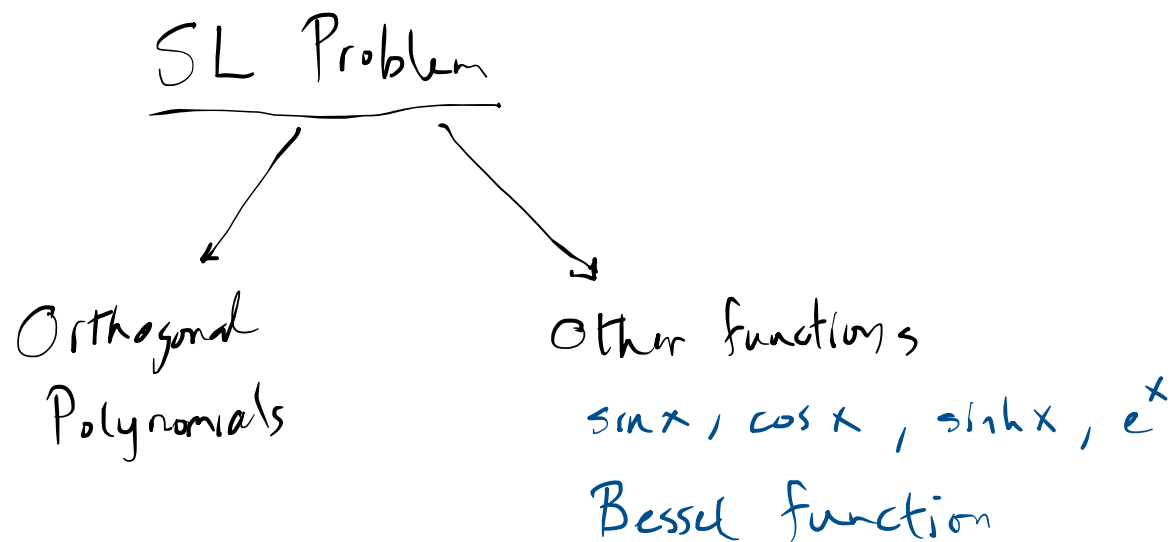


Ch14 Bessel Functions

Wednesday, April 8, 2026 4:45 PM



Bessel ODE $x^2 y'' + x y' + (x^2 - \nu^2) y = 0$

Frob. Method: $y = \sum_{m=0}^{\infty} a_m x^{m+s}$

indicial equation $s^2 - \nu^2 = 0 \Rightarrow s = \pm \nu$

$$s = \nu$$

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$$a_1 = 0$$

Rec. relation
$$a_{2m} = \frac{a_{2m-2}}{m(m+2\nu)}$$

1st kind
$$J_\nu(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{\Gamma(m+1)\Gamma(m+1+\nu)} \left(\frac{x}{2}\right)^{2m+\nu}$$

$s = -\nu$
$$J_{-\nu}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{\Gamma(m+1)\Gamma(m+1+\nu)} \left(\frac{x}{2}\right)^{2m-\nu}$$

For ν integer
$$J_{-\nu}(x) = (-1)^\nu J_\nu(x) \Rightarrow y_2(x) = \ln x \times J_\nu(x)$$

Bessel Function 2nd kind

Neumann (Weber) Function

2nd Kind

$$Y_\nu(x) = \frac{\cos(\pi\nu) J_\nu(x) - J_{-\nu}(x)}{\sin \pi\nu}$$

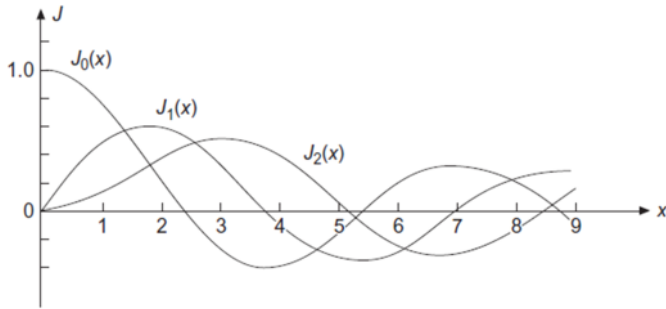


FIGURE 14.1 Bessel functions $J_0(x)$, $J_1(x)$, and $J_2(x)$.

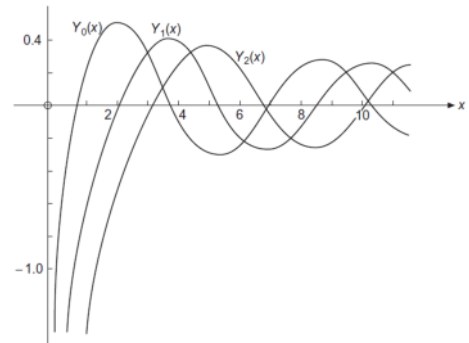


FIGURE 14.7 Neumann functions $Y_0(x)$, $Y_1(x)$, and $Y_2(x)$.

3rd kind Hankel Function

$$H_\nu^{(1)}(x) = J_\nu(x) + i Y_\nu(x)$$

$$H_\nu^{(2)}(x) = J_\nu(x) - i Y_\nu(x)$$

Spherical Bessel Function

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$$\nu = (2n+1)/2 = n + \frac{1}{2}$$

$$j_n(x) = \sqrt{\frac{\pi}{2x}} J_{(2n+1)/2}(x) = x^n \left(-\frac{1}{x} \frac{d}{dx} \right)^n \left(\frac{\sin x}{x} \right)$$

Rayleigh Formula

$$y_n(x) = \sqrt{\frac{\pi}{2x}} Y_{(2n+1)/2}(x) = -x^n \left(-\frac{1}{x} \frac{d}{dx} \right)^n \left(\frac{\cos x}{x} \right)$$

$$h_n^{(1)}(x) = j_n(x) + i y_n(x)$$

$$h_n^{(2)}(x) = j_n(x) - i y_n(x)$$

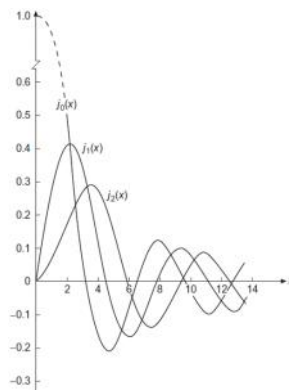


FIGURE 14.16 Spherical Bessel functions.

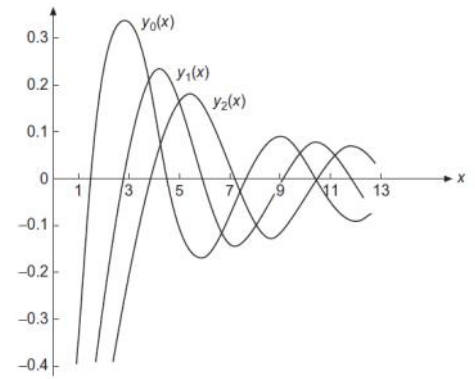


FIGURE 14.17 Spherical Neumann functions.

Modified Bessel

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$$x^2 y'' + xy' - (x^2 + \nu^2) y = 0$$

$$I_\nu(x) = i^{-\nu} J_\nu(ix)$$

$$K_\nu(x) = \frac{\pi}{2} i^{\nu+1} H_\nu^{(1)}(ix)$$

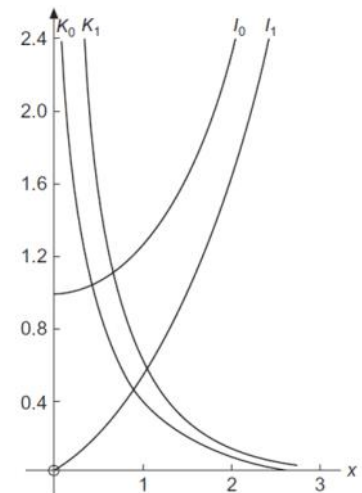


FIGURE 14.12 Modified Bessel functions.

<i>Function</i>	<i>Small x</i>	<i>Large x (asymptotic formulas)</i>
$J_p(x)$	$\frac{1}{\Gamma(p+1)} \left(\frac{x}{2}\right)^p + O(x^{p+2})$	$\sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{2p+1}{4}\pi\right) + O(x^{-3/2})$
$Y_p(x)$	$\begin{cases} p=0 & \frac{2}{\pi} \ln x + O(1) \\ p>0 & -\frac{\Gamma(p)}{\pi} \left(\frac{2}{x}\right)^p + \begin{cases} O(x^p), & p < 1 \\ O(x \ln \frac{1}{x}), & p = 1 \\ O(x^{2-p}), & p > 1 \end{cases} \end{cases}$	$\sqrt{\frac{2}{\pi x}} \sin\left(x - \frac{2p+1}{4}\pi\right) + O(x^{-3/2})$
$H_p^{(1) \text{ or } (2)}(x)$	Like $\pm i N_p(x)$	$\sqrt{\frac{2}{\pi x}} e^{\pm i[x - (2p+1)\pi/4]} + O(x^{-3/2})$
$I_p(x)$	Like $J_p(x)$	$\frac{1}{\sqrt{2\pi x}} e^x + O\left(\frac{e^x}{x}\right)$
$K_p(x)$	Like $-\frac{\pi}{2} N_p(x)$	$\sqrt{\frac{\pi}{2x}} e^{-x} + O\left(\frac{e^{-x}}{x}\right)$
$j_n(x)$	$\frac{x^n}{(2n+1)!!} + O(x^{n+2})$	$\frac{1}{x} \sin\left(x - \frac{n\pi}{2}\right) + O(x^{-2})$
$y_n(x)$	$\frac{-(2n-1)!!}{x^{n+1}} + O(x^{1-n})$	$-\frac{1}{x} \cos\left(x - \frac{n\pi}{2}\right) + O(x^{-2})$

14.1 Bessel Functions 1st kind

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$$J_\nu(x) = \sum_{s=0}^{\infty} \frac{(-1)^s}{s! \Gamma(\nu+s+1)} \left(\frac{x}{2}\right)^{\nu+2s}$$

$$g(x, t) = e^{\frac{x}{2}(t - \frac{1}{t})} = \sum_{n=-\infty}^{\infty} J_n(x) t^n$$

Recursion Relation

$$\frac{\partial g}{\partial t}$$

$$J_{n-1} - J_{n+1} = \frac{2n}{x} J_n$$

$$\frac{\partial g}{\partial x}$$

$$J_{n-1} - J_{n+1} = 2J'_n$$

$Z_\nu(x)$ set of function

$$x^2 Z_\nu'' + x Z_\nu' + (x^2 - \nu^2) Z_\nu = 0$$

$$x \rightarrow k\rho \Rightarrow \rho^2 \frac{d^2}{d\rho^2} Z_\nu(k\rho) + \rho \frac{d}{d\rho} Z_\nu(k\rho) + (\underline{k^2 \rho^2} - \nu^2) Z_\nu = 0$$

Integral Representation

$$\oint \frac{g(x,t)}{t^{n+1}} dt = 2\pi i J_n(x)$$

$$\text{let } t = e^{i\theta}$$

$$2\pi i J_n(x) = \int_0^{2\pi} \frac{1}{2\pi} e^{i(x \sin \theta - n\theta)} d\theta \quad \leftarrow$$

$$J_n(x) = \frac{1}{2\pi} \int_0^{2\pi} \cos(x \sin \theta - n\theta) d\theta$$

$$\Rightarrow \int_0^{2\pi} \sin(x \sin \theta - n\theta) d\theta = 0$$

$$J_0(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{ix \cos \theta} d\theta = \frac{1}{\pi} \int_0^{\pi} \cos(x \cos \theta) d\theta$$

14.2 Orthogonality

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Two functions: $\sin x$ and $\cos x$.

Consider just $\sin x$.

At the zeros of $\sin x$, namely,
 $x = n\pi$, $\sin x = 0$.

At $x = 1$, $\sin n\pi x = 0$.

The differential equation satisfied
 by $y = \sin n\pi x$ is
 $y'' + (n\pi)^2 y = 0$.

(In comparing the differential equations remember that p is a fixed constant. The correspondence is between the zeros of $\sin x$, namely $n\pi$, and the zeros of $J_p(x)$, namely, α, β , etc.)

We have proved (Chapter 7):

$$\int_0^1 \sin n\pi x \sin m\pi x dx = 0$$

for $n \neq m$.

Two functions for each p :
 $J_p(x)$ and $N_p(x)$.

Consider just $J_p(x)$ for one
 value of p .

At the zeros of $J_p(x)$, say
 $x = \alpha, \beta, \dots$, $J_p(x) = 0$.

At $x = 1$, $J_p(\alpha x) = 0$,
 $J_p(\beta x) = 0, \dots$

The differential equation satisfied
 by $y = J_p(\alpha x)$ is [see (16.5)]
 $x(xy')' + (\alpha^2 x^2 - p^2)y = 0$.

We shall prove:

$$\int_0^1 x J_p(\alpha x) J_p(\beta x) dx = 0$$

for $\alpha \neq \beta$.

$$\mathcal{L} J_\nu(k\rho) = k^2 J_\nu(k\rho)$$

$$\mathcal{L} = - \left(\frac{d^2}{d\rho^2} + \frac{1}{\rho} \frac{d}{d\rho} - \frac{\nu^2}{\rho^2} \right) = -\frac{1}{\rho} \left[\frac{d}{d\rho} \left(\rho \frac{d}{d\rho} \right) - \frac{\nu^2}{\rho} \right]$$

$$- \left[\frac{d}{d\rho} \left(\rho \frac{d}{d\rho} \right) - \frac{\nu^2}{\rho} \right] J_\nu = \underline{k^2 \rho J_\nu(k\rho)}$$

$$\rho \left[k' J_\nu(k\rho) J_\nu'(k'\rho) - k J_\nu'(k\rho) J_\nu(k'\rho) \right]_0^a = (k'^2 - k^2) \int_0^a \rho J_\nu(k\rho) J_\nu(k'\rho) d\rho$$

$$J_\nu(ka) = J_\nu(k'a) = 0 \quad ; \quad \alpha_{\nu,i} \equiv \text{ith zero of } J_\nu$$

$$\int_0^a \rho J_\nu(\alpha_{\nu,i} \frac{\rho}{a}) J_\nu(\alpha_{\nu,j} \frac{\rho}{a}) d\rho = 0 ; i \neq j$$

Normalization HW

$$\int_0^a \rho \left[J_\nu(\alpha_{\nu,i} \frac{\rho}{a}) \right]^2 d\rho =$$

$$\frac{a^2}{2} \left[J_{\nu+1}(\alpha_{\nu,i}) \right]^2$$

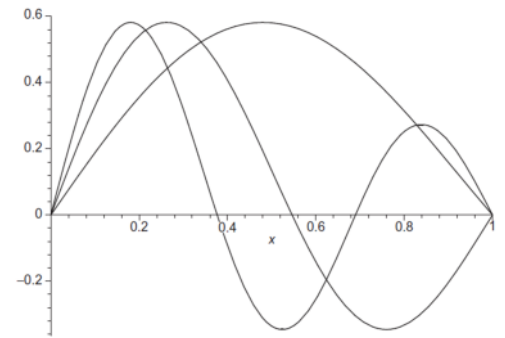


FIGURE 14.6 Bessel functions $J_1(\alpha_{1n}\rho)$, $n = 1, 2, 3$ on range $0 \leq \rho \leq 1$.

$$f(s) = \sum_j c_{vj} J_v(\alpha_{vj} \frac{s}{a}) \quad , \quad 0 \leq s \leq a$$

$$c_{vj} = \frac{2}{a^2 [J_{v+1}(\alpha_{vj})]^2} \int_0^a f(s) J_v(\alpha_{vj} \frac{s}{a}) s ds$$